National Testing Agency

Question Paper Name :	B Tech 4th Apr 2026 Shift 2
Subject Name :	B. Tech
Creation Date :	2026-04-04 19:40:25
Duration :	180
Total Marks :	300
Display Marks:	Yes

B. Tech

Group Number :	1
Group Id :	6952786
Group Maximum Duration :	0
Group Minimum Duration :	180
Show Attended Group? :	No
Edit Attended Group? :	No
Break time :	0
Group Marks :	300

Mathematics Section A

Section Id :	69527831
Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	20
Number of Questions to be attempted :	20
Section Marks :	80
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	69527831
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 1 Question Id : 695278376 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

For the function $f: [1,\infty) \to [1, \infty)$ defined by $f(x) = (x-1)^4 + 1$, among the two statements:

(I) The set $S = \{x \in [1, \infty) : f(x) = f^{-1}(x)\}$ contains exactly two elements, and

(II) The set $S = \{x \in [1, \infty) : f(x) = f^{-1}(x+1)\}$ is an empty set,

Options :

6952781276. only (I) is TRUE

6952781277. only (II) is TRUE

6952781278. both (I) and (II) are TRUE

6952781279. neither (I) nor (II) is TRUE

Question Number : 2 Question Id : 695278377 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $S = \{z \in \mathbb{C} : z^2 + 4z + 16 = 0\}$. Then $\sum_{z \in S} | z + \sqrt{3}\,i |^2$ is equal to:

Options :

6952781280. 42

6952781281. 23

6952781282. 27

6952781283. 38

Question Number : 3 Question Id : 695278378 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

If the system of equations:

$x + y + z = 5$

$x + 2y + 3z = 9$

$x + 3y + \lambda z = \mu$

has infinitely many solutions, then the value of $\lambda + \mu$ is:

Options :

6952781284. 16

6952781285. 18

6952781286. 19

6952781287. 21

Question Number : 4 Question Id : 695278379 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

If $\alpha = 1$ and $\beta = 1 + i\sqrt{2}$, where $i = \sqrt{-1}$ are two roots of the equation

$x^3 + ax^2 + bx + c = 0$, $a, b, c \in \mathbb{R}$, then $\int_{-1}^{1} \left(x^3 + ax^2 + bx + c\right) dx$ is equal to:

Options :

6952781288. -2

6952781289. -4

6952781290. -8

6952781291. -10

Question Number : 5 Question Id : 695278380 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

If the quadratic equation $(\lambda + 2)x^2 - 3\lambda x + 4\lambda = 0$, $\lambda \neq -2$, has two positive roots, then the number of possible integral values of λ is:

Options :

6952781292. 1

6952781293. 2

6952781294. 3

6952781295. 4

Question Number : 6 Question Id : 695278381 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $A=\begin{bmatrix}1 & 2 & 7\\ 4 & -2 & 8\\ 3 & 8 & -7\end{bmatrix}$ and $\det(A-\alpha I)=0$, where α is a real number. If the largest possible value of α is p, then the circle $(x-p)^2+(y-2p)^2=320$, intersects the co-ordinate axes at

Options :

6952781296. 1 point

6952781297. 2 points

6952781298. 3 points

6952781299. 4 points

Question Number : 7 Question Id : 695278382 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $\alpha=\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+\ldots\infty$ and $\beta=\frac{1}{3}+\frac{1}{9}+\frac{1}{27}+\ldots\infty$. Then the value of $(0.2)^{\log_{\sqrt{5}}(\alpha)}+(0.04)^{\log_5(\beta)}$ is equal to:

Options :

6952781300. 4

6952781301. 5

6952781302. 8

6952781303. 25

Question Number : 8 Question Id : 695278383 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

For 10 observations $x_1, x_2, \ldots, x_{10}$, if $\sum_{i=1}^{10}(x_i+2)^2=180$ and $\sum_{i=1}^{10}(x_i-1)^2=90$, then their standard deviation is:

Options :

6952781304. 2

6952781305. $\sqrt{3}$

6952781306. $2\sqrt{2}$

6952781307. 3

Question Number : 9 Question Id : 695278384 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

In the expansion of $\left(9x-\frac{1}{3\sqrt{x}}\right)^{18}$, $x>0$, if the term independent of x is (221)k, then k is equal to:

Options :

6952781308. 84

6952781309. 78

6952781310. 168

6952781311. 198

Question Number : 10 Question Id : 695278385 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let P (3cos α, 2sin α), $\alpha \neq 0$, be a point on the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$, Q be a point on the circle $x^2+y^2-14x-14y+82=0$ and R be a point on the line $x+y=5$ such that the centroid of the triangle PQR is $\left(2+\cos\alpha, 3+\frac{2}{3}\sin\alpha\right)$. Then the sum of the ordinates of all possible points R is:

Options :

6952781312. 6

6952781313. 2

6952781314. 4

6952781315. 8

Question Number : 11 Question Id : 695278386 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $H: \frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ be a hyperbola such that the distance between its foci is 6 and the distance between its directrices is $\frac{8}{3}$. If the line $x=\alpha$ intersects the hyperbola H at the points A and B such that the area of the triangle AOB is $4\sqrt{15}$, where O is the origin, then α^2 equals

Options :

6952781316. 12

6952781317. 16

6952781318. 24

6952781319. 25

Question Number : 12 Question Id : 695278387 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

$\max\limits_{0 \le x \le \pi}\left(16\sin\left(\frac{x}{2}\right)\cos^3\left(\frac{x}{2}\right)\right)$ is equal to:

Options :

6952781320. $\frac{3\sqrt{3}}{2}$

6952781321. $3\sqrt{3}$

6952781322. $4\sqrt{3}$

6952781323. $6\sqrt{3}$

Question Number : 13 Question Id : 695278388 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The shortest distance between the lines

$\vec{r} = \left(\frac{1}{3}\hat{i} + 2\hat{j} + \frac{8}{3}\hat{k}\right) + \lambda\left(2\hat{i} - 5\hat{j} + 6\hat{k}\right)$

and $\vec{r} = \left(-\frac{2}{3}\hat{i} - \frac{1}{3}\hat{k}\right) + \mu\left(\hat{j} - \hat{k}\right), \lambda, \mu \in \mathbb{R}$, is:

Options :

6952781324. $\sqrt{5}$

6952781325. 3

6952781326. $2\sqrt{3}$

6952781327. $\sqrt{15}$

Question Number : 14 Question Id : 695278389 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

If $\left(2\alpha + 1, \alpha^2 - 3\alpha, \frac{\alpha - 1}{2}\right)$ is the image of $(\alpha, 2\alpha, 1)$ in the line

$\frac{x-2}{3} = \frac{y-1}{2} = \frac{z}{1}$, then the possible value(s) of α is (are)

Options :

6952781328. Only 3

6952781329. Only 3 and -1

6952781330. Only 3, $\frac{1}{4}$ and -1

6952781331. Only 3 and $\frac{1}{4}$

Question Number : 15 Question Id : 695278390 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $\hat{u}$ and $\hat{v}$ be unit vectors inclined at an acute angle such that $\left|\hat{u}\times\hat{v}\right|=\frac{\sqrt{3}}{2}$. If $\vec{A}=\lambda\,\hat{u}+\hat{v}+\left(\hat{u}\times\hat{v}\right)$, then λ is equal to:

Options :

6952781332. $\frac{4}{3}\left(\vec{A}\cdot\hat{u}\right)-\frac{2}{3}\left(\vec{A}\cdot\hat{v}\right)$

6952781333. $\frac{2}{3}\left(\vec{A}\cdot\hat{u}\right)-\frac{1}{3}\left(\vec{A}\cdot\hat{v}\right)$

6952781334. $\frac{4}{3}\left(\vec{A}\cdot\hat{u}\right)+\frac{2}{3}\left(\vec{A}\cdot\hat{v}\right)$

6952781335. $\left(\vec{A}\cdot\hat{u}\right)-\frac{1}{2}\left(\vec{A}\cdot\hat{v}\right)$

Question Number : 16 Question Id : 695278391 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let for some $\alpha\in\mathbb{R}$, $f:\mathbb{R}\to\mathbb{R}$ be a function satisfying $f(x+y)=f(x)+2y^2+y+\alpha xy$ for all $x, y\in\mathbb{R}$. If $f(0)=-1$ and $f(1)=2$, then the value of $\sum_{n=1}^{5}\left(\alpha+f\left(n\right)\right)$ is:

Options :

6952781336. 110

6952781337. 140

6952781338. 150

6952781339. 170

Question Number : 17 Question Id : 695278392 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let

$A=\{(a, b, c) : a, b, c$ are non-negative integers and $a+b+2c=22\}$.

Then $n(A)$ is equal to:

Options :

6952781340. 121

6952781341. 124

6952781342. 144

6952781343. 169

Question Number : 18 Question Id : 695278393 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The area of the region bounded by the curves $x + 3y^2 = 0$ and $x + 4y^2 = 1$ is equal to:

Options :

6952781344. $\frac{1}{3}$

6952781345. $\frac{2}{3}$

6952781346. $\frac{4}{3}$

6952781347. $\frac{5}{3}$

Question Number : 19 Question Id : 695278394 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $y = y(x)$ be the solution of the differential equation:

$$\frac{dy}{dx} + \left(\frac{6x^2 + \left(3x^2 + 2x^3 + 4\right)e^{-2x}}{\left(x^3 + 2\right)\left(2 + e^{-2x}\right)} \right) y = 2 + e^{-2x},$$

$x \in (-1, 2)$, satisfying $y(0) = \frac{3}{2}$. If $y(1) = \alpha(2 + e^{-2})$, then α is equal to:

Options :

6952781348. $\frac{13}{8}$

6952781349. $\frac{6}{13}$

6952781350. $\frac{12}{13}$

6952781351. $\frac{13}{12}$

Question Number : 20 Question Id : 695278395 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The integral $\int_0^1 \cot^{-1}\left(1 + x + x^2\right) dx$ is equal to:

Options :

6952781352. $2\tan^{-1}2+\frac{1}{2}\log_e\left(\frac{5}{4}\right)+\frac{\pi}{2}$

6952781353. $2\tan^{-1}2+\frac{1}{2}\log_e\left(\frac{5}{4}\right)-\frac{\pi}{2}$

6952781354. $2\tan^{-1}2-\frac{1}{2}\log_e\left(\frac{5}{4}\right)+\frac{\pi}{2}$

6952781355. $2\tan^{-1}2-\frac{1}{2}\log_e\left(\frac{5}{4}\right)-\frac{\pi}{2}$

Mathematics Section B

Section Id :	69527832
Section Number :	2
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	5
Number of Questions to be attempted :	5
Section Marks :	20
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	69527832
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 21 Question Id : 695278396 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

From a month of 31 days, 3 different dates are selected at random. If the probability that these dates are in an increasing A.P. is equal to $\frac{a}{b}$, where $a, b \in$ N and $gcd\,(a, b) = 1$, then $a + b$ is equal to _____

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 22 Question Id : 695278397 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Let $f(x)=\begin{cases} e^{x-1} & , x<0 \\ x^2-5x+6 & , x\geq 0 \end{cases}$ and $g(x)=f(|x|)+|f(x)|$. If the number of points where g is not continuous and is not differentiable are α and β respectively, then $\alpha + \beta$ is equal to _____

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal

Text Areas : PlainText
Possible Answers :
1

Question Number : 23 Question Id : 695278398 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Let A, B be points on the two half-lines $x - \sqrt{3}|y| = \alpha$, $\alpha > 0$ at a distance of α from their point of intersection P. The line segment AB meets the angle bisector of the given half-lines at the point Q. If $PQ = \frac{9}{2}$ and R is the radius of the circumcircle of ΔPAB, then $\frac{\alpha^2}{R}$ is equal to _____

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 24 Question Id : 695278399 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Let A, B and C be the vertices of a variable right angled triangle inscribed in the parabola $y^2 = 16x$. Let the vertex B containing the right angle be (4, 8) and the locus of the centroid of ΔABC be a conic C_o. Then three times the length of latus rectum of C_o is _____

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 25 Question Id : 695278400 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Let f be a twice differentiable function such that

$$f(x) = \int_0^x \tan(t-x)\,dt - \int_0^x f(t)\tan t\,dt,\ x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right).$$

Then $f''\left(\frac{\pi}{6}\right) + 12\,f'\left(-\frac{\pi}{6}\right) + f\left(\frac{\pi}{6}\right)$ is equal to _____

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Physics Section A

Section Id : 69527833
Section Number : 3